CRITERION FOR A QUOTIENT STACK TO BE ALGEBRAIC

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span; proof complete as written.

Allowed standard prerequisites (editorial).

- Restriction equivalence. spaces-groupoids-lemma-quotient-stack-restrict-equivalence (cross-chapter, named). [U'/R'] -> [U/R] is an equivalence under (1).
- Flat groupoid theorem. theorem-flat-groupoid-gives-algebraic-stack (same chapter, named). Algebraicity of [U'/R'].
- Quotient stack objects and morphisms. spaces-groupoids-lemma-quotient-stack-objects, spaces-groupoids-lemma-quotient-stack-morphisms (cross-chapter, named). Explicit description of the quotient stack over W.
- Essential surjectivity criterion. stacks-lemma-characterize-essentially-surjective-when-ff (cross-chapter, named). Fully faithful plus local essential surjectivity.

Source reference. Original reference 045C: focused-seeds/geometry/S9-crit-quotient-algebraic/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 045B: focused-seeds/geometry/S9-crit-quotient-algebraic/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03YQ: focused-seeds/geometry/S9-crit-quotient-algebraic/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04T4: focused-seeds/geometry/S9-crit-quotient-algebraic/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04WY: focused-seeds/geometry/S9-crit-quotient-algebraic/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof

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Source reference. Original reference 04SS: focused-seeds/geometry/S9-crit-quotient-algebraic/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 044V: focused-seeds/geometry/S9-crit-quotient-algebraic/ref-tags.txt, source lines 11–11. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 044X: focused-seeds/geometry/S9-crit-quotient-algebraic/ref-tags.txt, source lines 12–12. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 046T: focused-seeds/geometry/S9-crit-quotient-algebraic/ref-tags.txt, source lines 13–13. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 046N: focused-seeds/geometry/S9-crit-quotient-algebraic/ref-tags.txt, source lines 14–14. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original theorem 06FI: sources/stacks-criteria/pinned-original-source.tex, source lines 3245–3245. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme and let B be an algebraic space over S. Let (U, R, s, t, c) be a groupoid in algebraic spaces over B. The quotient stack $[U/R]$ is an algebraic stack if and only if there exists a morphism of algebraic spaces $g : U' \to U$ such that*

1. *the composition $U' \times_{g,U,t} R \to R \xrightarrow{s} U$ is a surjection of sheaves, and*
2. *the morphisms $s', t' : R' \to U'$ are flat and locally of finite presentation where (U', R', s', t', c') is the restriction of (U, R, s, t, c) via g.*

Proof. First, assume that $g : U' \to U$ satisfies (1) and (2). Property (1) implies that $[U'/R'] \to [U/R]$ is an equivalence, see Groupoids in Spaces, Lemma 046T. By Theorem 06FI the quotient stack $[U'/R']$ is an algebraic stack. Hence $[U/R]$ is an algebraic stack too, see Algebraic Stacks, Lemma 03YQ.

Conversely, assume that $[U/R]$ is an algebraic stack. We may choose a scheme W and a surjective smooth 1-morphism

$$f : (Sch/W)_{fppf} \longrightarrow [U/R].$$

By the 2-Yoneda lemma (Algebraic Stacks, Section 04SS) this corresponds to an object ξ of $[U/R]$ over W. By the description of $[U/R]$ in Groupoids in Spaces, Lemma 044X we can find a surjective, flat, locally finitely presented morphism $b : U' \to W$ of schemes such that $\xi' = b^*\xi$ corresponds to a morphism $g : U' \to U$. Note that the 1-morphism

$$f' : (Sch/U')_{fppf} \longrightarrow [U/R].$$

corresponding to ξ' is surjective, flat, and locally of finite presentation, see Algebraic Stacks, Lemma 045B. Hence $(Sch/U')_{fppf} \times_{[U/R]} (Sch/U')_{fppf}$ which is represented by the algebraic space

$$Isom_{[U/R]}(\mathrm{pr}_0^*\xi', \mathrm{pr}_1^*\xi') = (U' \times_S U') \times_{(g\circ\mathrm{pr}_0, g\circ\mathrm{pr}_1), U\times_S U} R = R'$$

(see Groupoids in Spaces, Lemma 044V for the first equality; the second is the definition of restriction) is flat and locally of finite presentation over U' via both s' and t' (by base change, see Algebraic Stacks, Lemma 045C). By this description of R' and by Algebraic Stacks, Lemma 04T4 we obtain a canonical fully faithful 1-morphism $[U'/R'] \to [U/R]$. This 1-morphism is essentially surjective because f' is flat, locally of finite presentation, and surjective (see Stacks, Lemma 046N); another way to prove this is to use Algebraic Stacks, Remark 04WY. Finally, we can use Groupoids in Spaces, Lemma 046T to conclude that the composition $U' \times_{g,U,t} R \to R \xrightarrow{s} U$ is a surjection of sheaves. □